

Deviance

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EDMs and GLMs

We say that a distribution belongs to the **exponential dispersion model** family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$\begin{aligned} Y_i | \mathbf{x}_i &\sim \text{EDM}(\mu_i, \phi) \\ g(\mu_i) &= \mathbf{x}_i^T \boldsymbol{\beta} \end{aligned}$$

Notice: We want to model the mean μ of an EDM, but the probability function is written in terms of θ .

Rewriting the probability function

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Unit deviance and dispersion model form

Unit deviance: $d(y, \mu) = 2[t(y, y) - t(y, \mu)]$

Example: Normal

Example: Bernoulli

Why do we care?

Activity

Work on the activity (handout).